

Chapter 7 Warm-Up: Importing Functions from Modules

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1 Creating a Module

1. A Python **module** is a .py file containing code that can be imported into another program.
Create a new file called

`math_tools.py`

and place the following functions inside it:

```
def square(x):  
    return x * x  
  
def cube(x):  
    return x * x * x  
  
def average(x, y):  
    return (x + y) / 2
```

Save the file. The functions are now organized inside the module `math_tools`.

2 Uploading to Google Colab

1. Open a new Google Colab notebook.
Upload `math_tools.py` to the Colab runtime:
 1. Click the **Files** folder icon on the left side.
 2. Click the **Upload** icon in the Files panel.
 3. Select `math_tools.py` from your computer.
 4. Confirm that `math_tools.py` appears in the Colab file list.The module must be in the current Colab runtime before Python can import it.

3 Importing the Module

1. In a Colab code cell, import the complete module:

```
import math_tools
```

Functions inside the module are accessed using the module name followed by a period:

```
print(math_tools.square(5))  
print(math_tools.cube(3))  
print(math_tools.average(8, 12))
```

Before running the cell, predict the three outputs.

- What does the name `math_tools` refer to?
- Which part of
`math_tools.square(5)`
is the module name? Which part is the function name?

4 Importing a Function

1. Python can also import an individual function:

```
from math_tools import square
```

Now call it without the module name:

```
print(square(7))
```

Compare this with

```
import math_tools  
print(math_tools.square(7))
```

Both call the same function, but they use different import statements.

5 Code-Lab Exercise

1. Add the following function to `math_tools.py`:

```
def classify_score(score):  
    if score >= 4:  
        return "Calculus"  
    elif score >= 2:  
        return "Trigonometry"  
    else:  
        return "Algebra"
```

Upload the updated file to Colab. Then import the function:

```
from math_tools import classify_score
```

Test it using

```
scores = [0, 2, 3, 5]
```

```
for score in scores:  
    course = classify_score(score)  
    print(score, course)
```

- Predict the four course assignments before running the code.
- Which file **defines** the function?
- Which file or notebook **calls** the function?
- Add one more score to the list and verify that the imported function still performs the placement correctly.

6 Important

Files uploaded directly to a Colab runtime are temporary. If the runtime is restarted or disconnected, you may need to upload your module again.